University of St.Gallen

# CryptoFlow Event-Driven & Process-Oriented Crypto Portfolio Platform

EDPO Project FS26

St.Gallen | Ioannis Theodosiadis & Cyril Gabriele

From insight to impact.

# Domain Knowledge needed for CryptoFlow

From the perspective of the buyer:

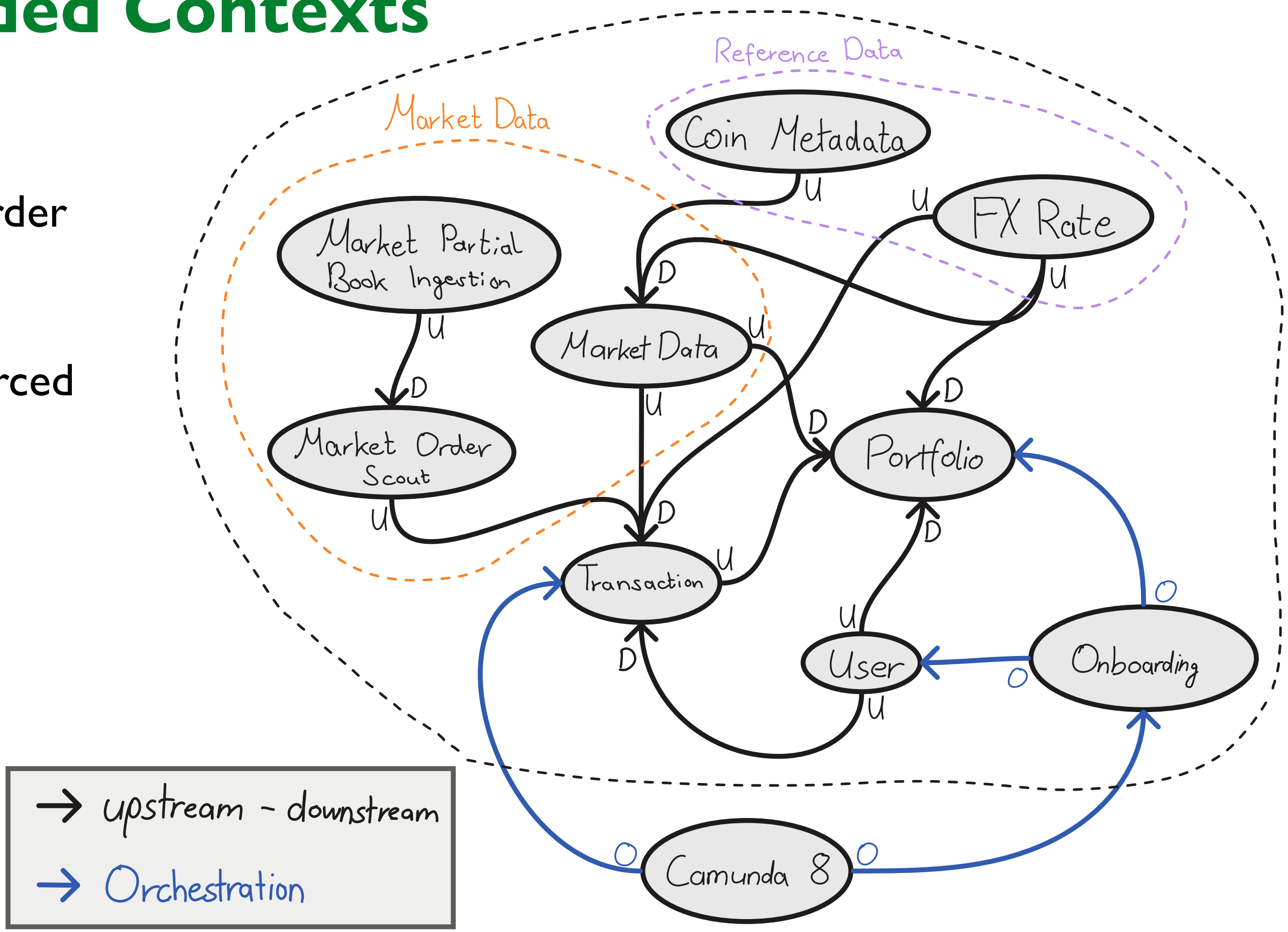
- Bid: How much am I willing to pay?
- Ask: How much does the other person demand?
- Portfolio: How much do I own? E.g., 1 BTCUSDT

---

- Order Book: Contains the matches of Bids and Asks
- FX-Rate: Foreign exchange rate (currency)
- OHCL: Open, high, low, close is a standard for illustrating price movement over a specific period e.g., candlestick graphs

# Domain: **Nine Bounded Contexts**

- **Market Data**
  - Live market feeds, partial order book, OHLC diagram
- **Reference Data**
  - Slow-moving externally sourced facts for event enrichment
- **Portfolio**
  - Holdings & valuation
- **Trading**
  - Order lifecycle
- **User**
  - Accounts & confirmation
- **Onboarding**
  - Saga orchestrator



# Concepts – **What Concepts Did We Implement?**

- **Stateless processing**
  - Three producer pipelines for fx-rate, coin-metadata, and order-book with event-filtering, event-routing, event-translator.
- **Stateful processing**
  - Persistent local state stores track pending bids, matched transactions, allocated asks, three window stores for OHLC (1m/5m/1h), four KTables (holdings, prices, position-value, portfolio-value), and dashboard stats. All changelog-replicated for fault-tolerance.
- **Interactive queries**
  - Portfolio value and market scout read directly from local state store instead of consuming output topics.
- **Custom matching**
  - Bids and asks are rekeyed by symbol, merged, and matched with a custom stateful transformer.

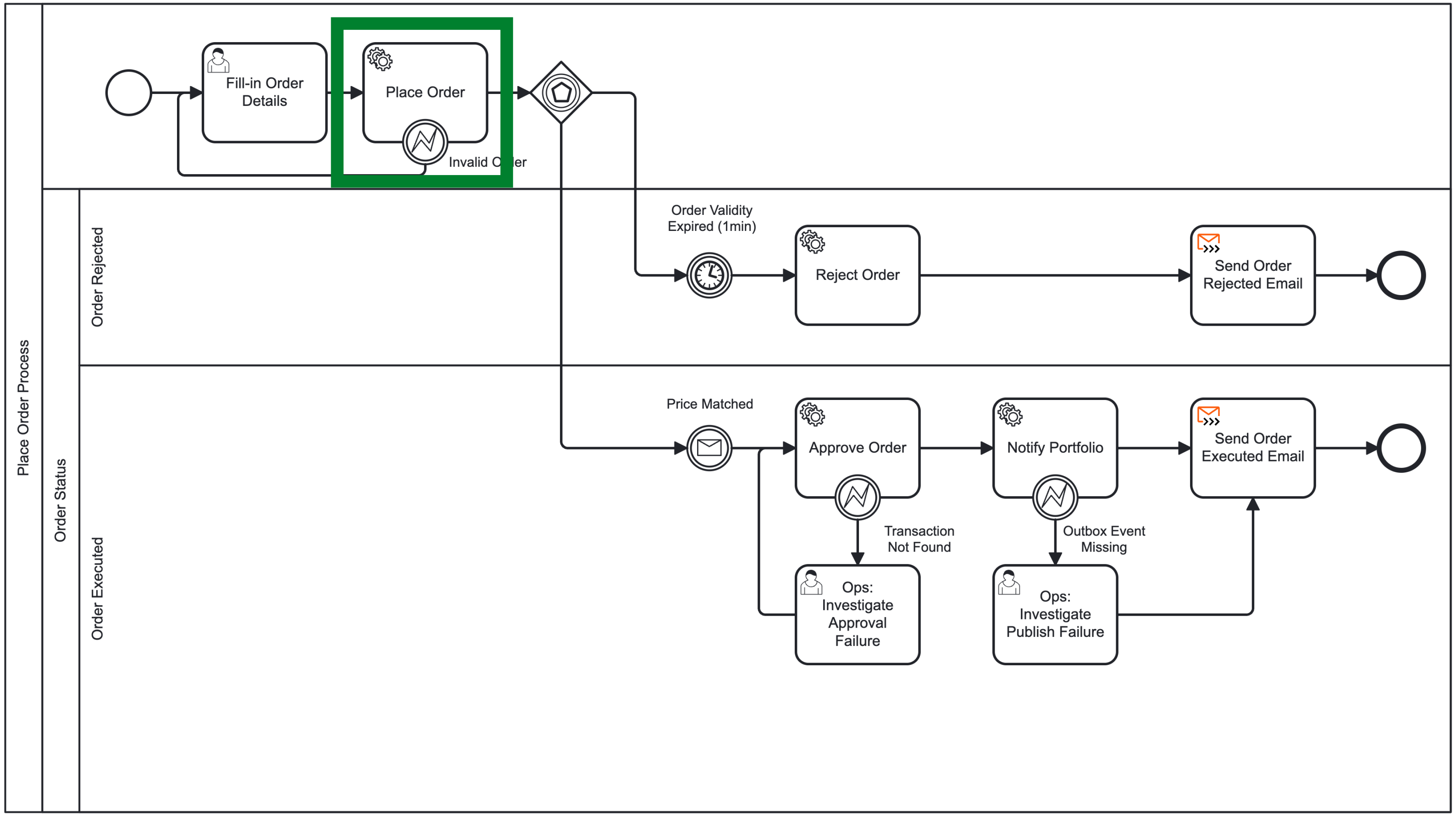
# Concepts – **What Concepts Did We Implement?**

- **Contracts**
  - JSON for boundary/raw topics. Registryless Avro where one team owns producer and consumer. Registered Avro with schema registry where contracts span cross service.
- **Multiphase Repartitioning**
  - Portfolio valuation runs in parallel over all partitions looking for FK (userId, symbol), then re-keys events to a new partition with key = userId to sum them up.
- **Reprocessing**
  - Deterministically rebuilds state stores from input topics from offset 0 for bug fixes and schema evolution.
- **Compacted Topic as cache**
  - FX-rates, coin-metadata, user-currency, portfolio-value are materialised as KTables/GlobalKTables in every consumer, enabling reads from process memory.

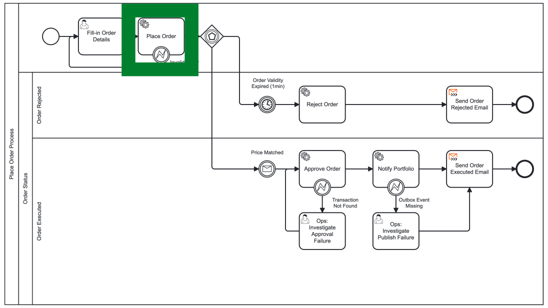
# Concepts – What Concepts Did We Implement?

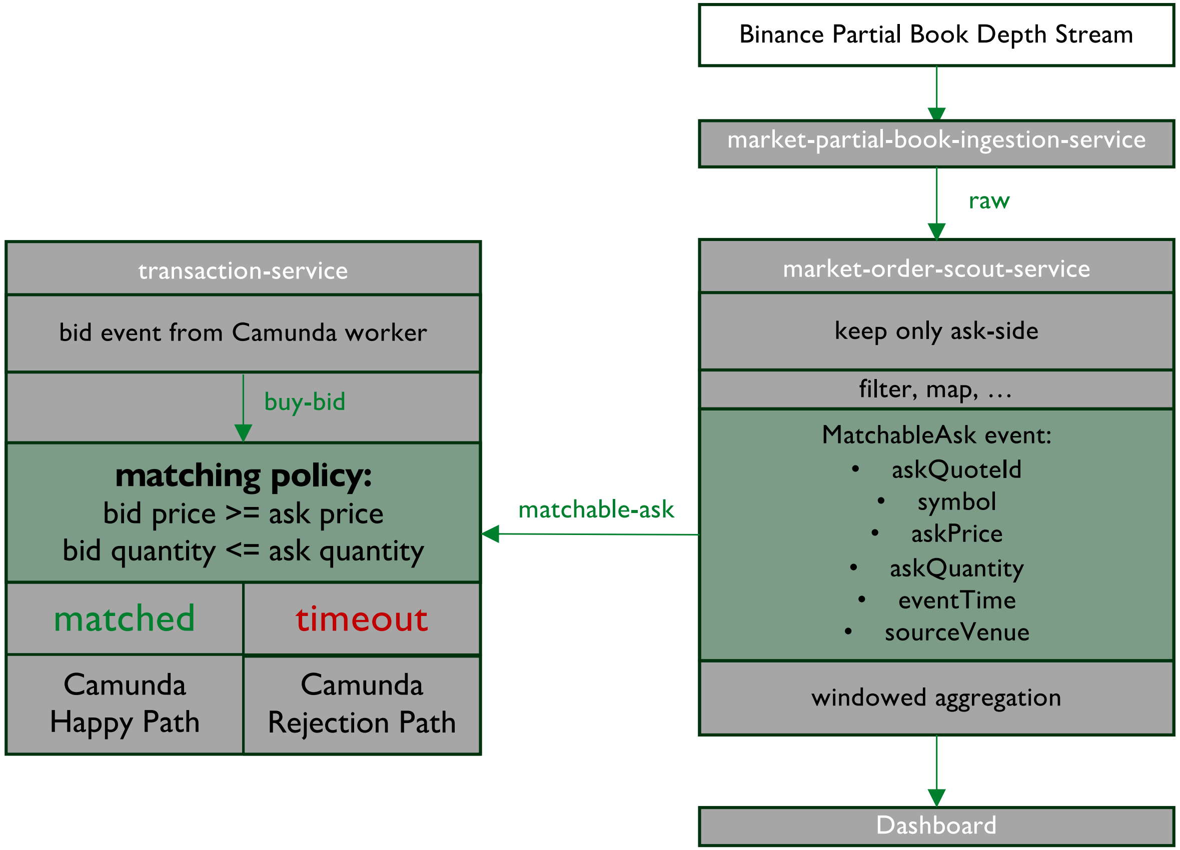
- **Windowing**
  - Tumbling event-time windows aggregate ask opportunities and minimum ask prices per symbol.
  - Sliding join windows handle concurrent bids on ask opportunities and close on the earlier event.
  - Three OHLC pipelines aggregate into closed candlestick bars.
- **Custom Timestamp Extractor**
  - Payload event time oversees window assignment and matching in order matching and OHLC.
- **Grace**
  - Grace periods absorb late events for OHLC.
  - Late asks can still match while bids are retained for validityWindow + gracePeriod.
- **Suppress**
  - OHLC calls suppress until window closes so each event emitted in one of the three topics (1m/5m/1h) is a single, finalized closed bar.
- **Joins**
  - KTable x KTable for portfolio valuation. KStream x KStream in bid/ask matcher. KStream x GlobalKTable lookup enriches market data with coin metadata.

# Place Order Process – **Enhancement with Kafka**

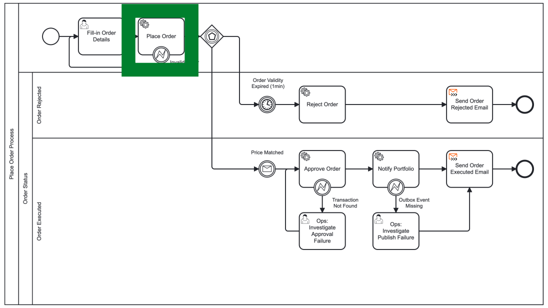


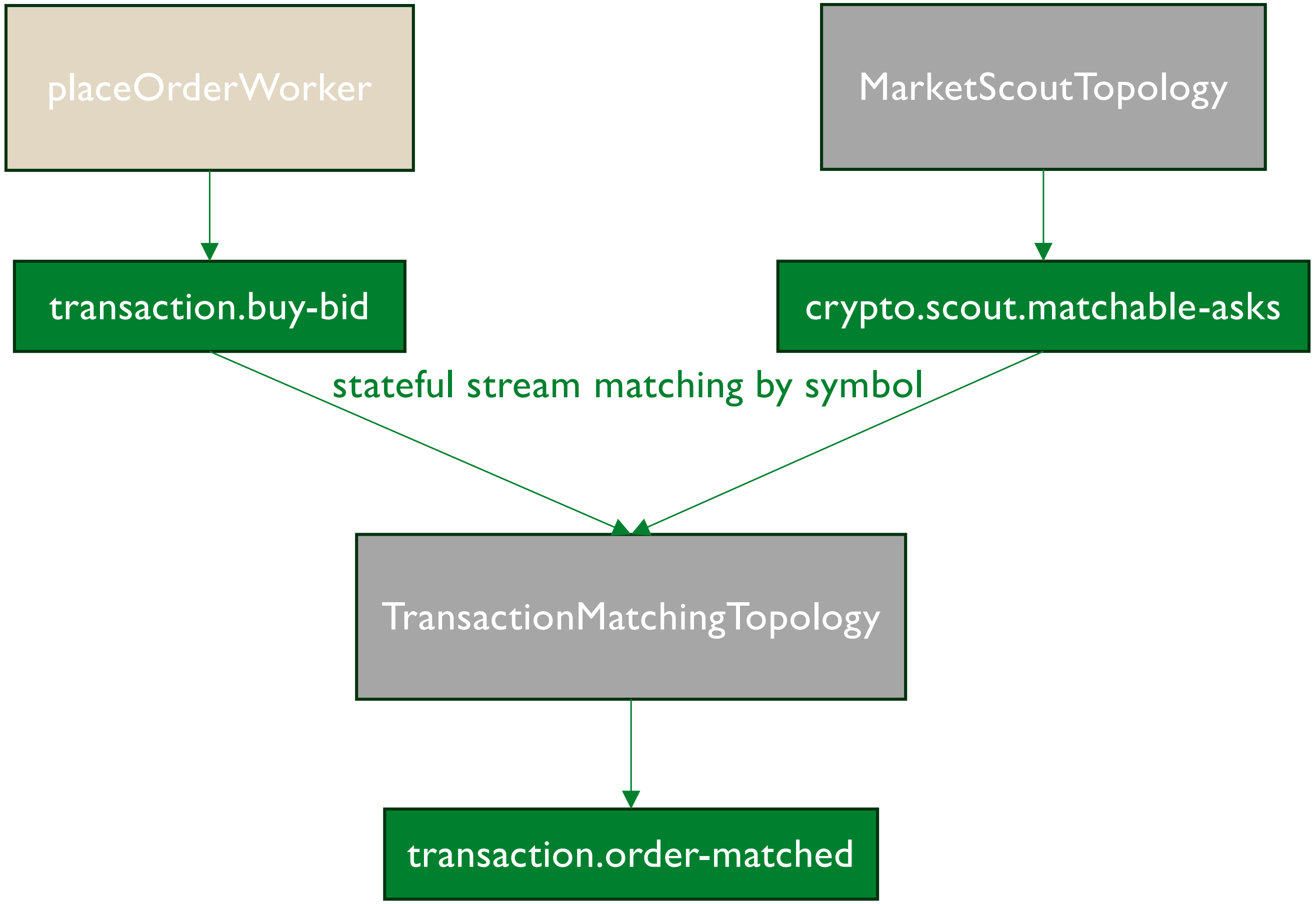
# Place Order Process – **Enhancement with Kafka**



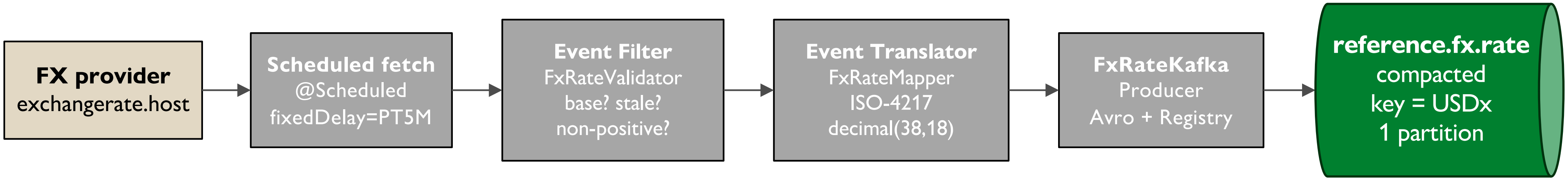



# Place Order Process – **Enhancement with Kafka**



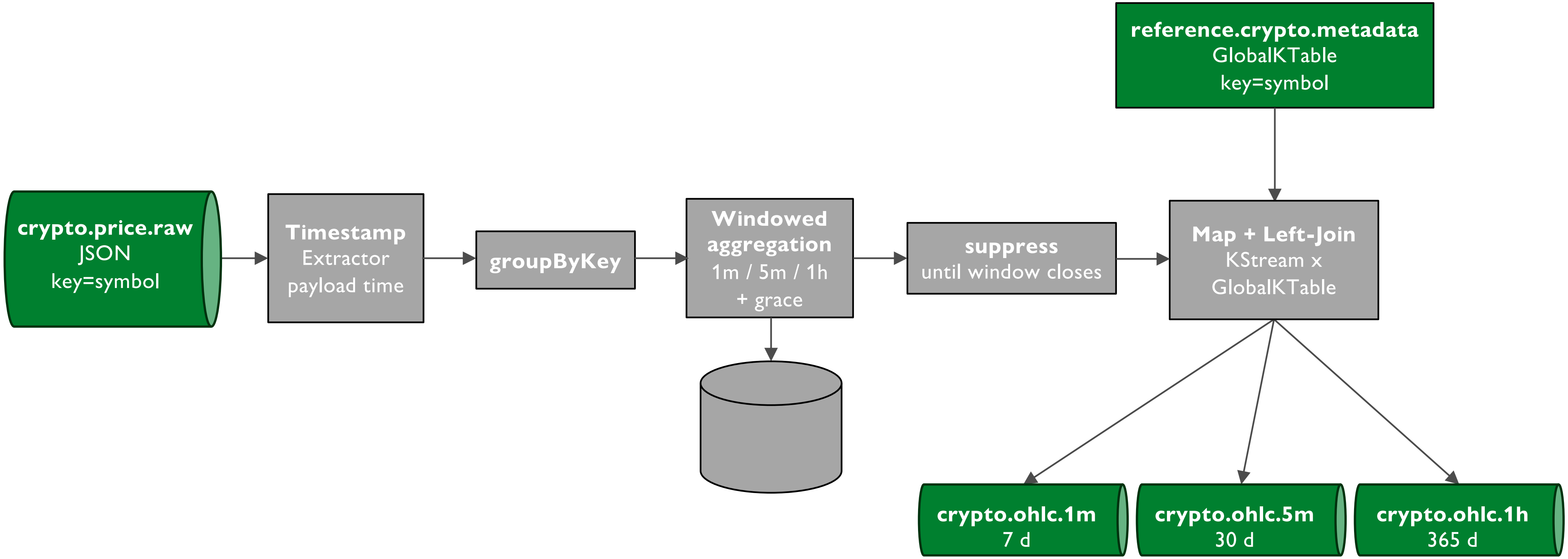


# Topology: **Exchange Rate**

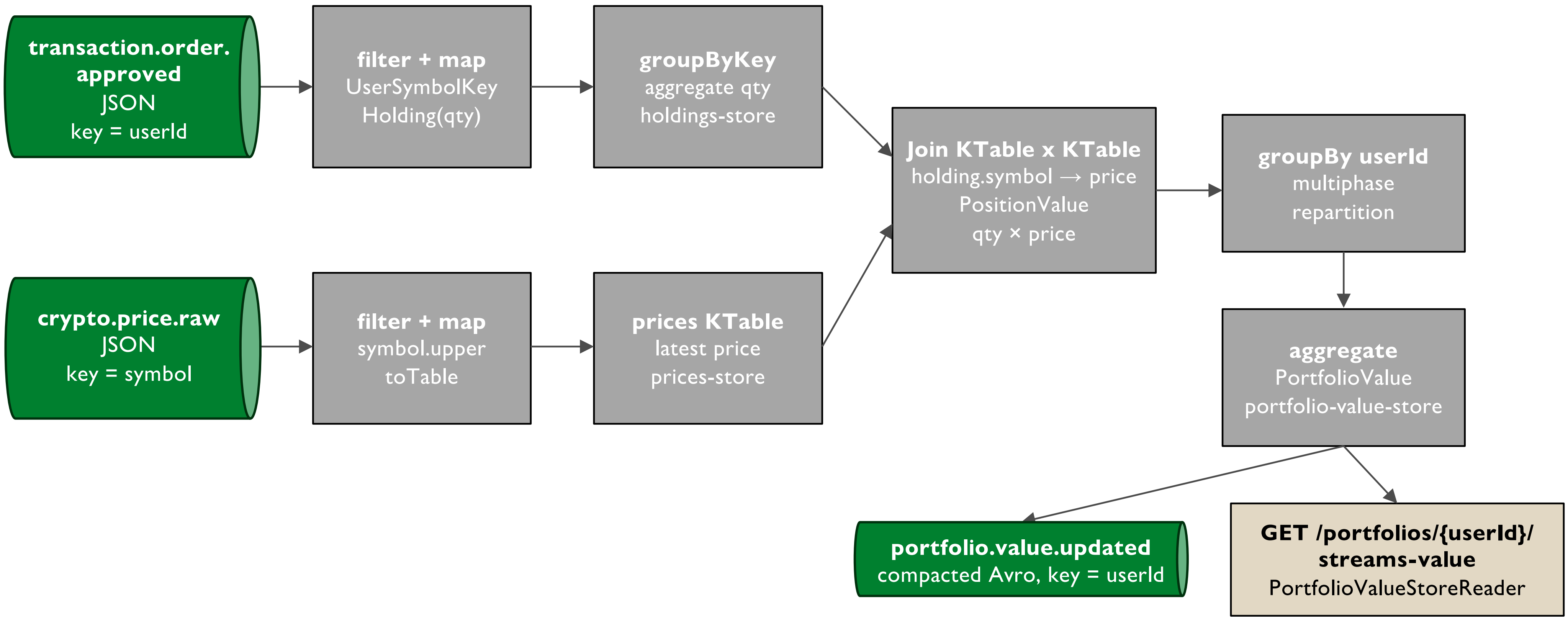



Filter + Translator → compacted KTable cache

# Topology: **OHLC Streams**



# Topology: **Portfolio Valuation**




FK join → repartition → local state + Interactive Queries

# Insights: **Challenges & Lessons Learned**

- Compacted topics can act as caches.
  - Used four times in total, but cache lives on broker
- Event splitting produces a lot of data.
  - Reconfigure cleanup policy, lower retention period, increase partition size, but old events are disregarded very quickly

- Independent, single topologies are easily constructed. Topologies using joins, mixing stateless, stateful and windowing concepts become very complex.
  - Producing a clean design for matching bids and asks was highly demanding.
- Once the actual process is discussed and the topology is drawn, the implementation becomes a matter of minutes.

# Demonstration

University of St.Gallen

# CryptoFlow
# Event-Driven & Process-Oriented Crypto Portfolio Platform

EDPO Project FS26

St.Gallen | Ioannis Theodosiadis & Cyril Gabriele

From insight to impact.